



UTM

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PROJECT PROPOSAL

THEME: HEALTHCARE

(MINDCARE AI - INTELLIGENT MENTAL HEALTH SUPPORT CHATBOT)

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TABLE OF CONTENTS

1.0 INTRODUCTION.....	3
PROBLEM STATEMENT.....	4
2.0 AI SOLUTION.....	5
3.0 THE GOAL OF AI SOLUTION.....	5
4.0 PROCESS OF EMPATHIZE IN DESIGN THINKING.....	6
WHO are we empathizing with?.....	6
What do they need to DO?.....	6
What do they SEE?.....	6
What do they SAY?.....	6
What do they DO?.....	7
What do they HEAR?.....	7
What do they THINK AND FEEL?.....	7
5.0 PROCESS OF DEFINE IN DESIGN THINKING.....	7

1.0 INTRODUCTION

In modern society, mental health appears to take a center stage. People of different demographics and backgrounds, such as students, working individuals, parents, elderly, and many others, are undergoing more stress, anxiety, and burnout than they have ever had. However, despite the fact that mental-health problems are no longer neglected, a considerable amount of victims do not seek help because of stigma, the lack of access to care, high cost, lengthy waits, or fear of being judged.

The recent developments in the area of artificial intelligence (AI) and other technologies provide significant opportunities with the redefinition of providing mental-health assistance. AI-based solutions may provide prompt, easily accessible, and judgment-free support to supplement the traditional mental-health services. Additionally, these technologies can be used to detect trends of emotional distress, provide real-time coping mechanisms, and refer users to the relevant resources and support privacy and confidentiality.

Given this, we have suggested MindCare AI, which is a smart chatbot meant to serve as mental-health assistance to the general population. The system uses natural language processing and sentiment analysis to identify the emotional tone of user communications and produces empathetic and personalized responses with appropriate mental-health resources. MindCare AI is an affordable first-line support channel, available 24/7 to listen to people, maintain and lead them through everyday stressors, and refer to the professional care where required.

PROBLEM STATEMENT

1. People have no idea where they can get mental-health help and are usually ignorant about the right course of action in the moments of stress, anxiety, or emotional disturbance.
2. The lack of instant support increases the problem of dealing with stress and emotions, especially in one day or on weekends or other stressful situations that can induce crises.
3. Stigma and the fear of judgement are both further reasons that do not allow people to seek help and leads to silent suffering, which does not allow any interaction with counselors, therapies, or support groups.
4. Mental-health problems are often unspotted until they grow big and destroy chances of early treatment and assistance.
5. The lack of mental-health services cannot meet the great demand in communities which results in long waiting lines, high prices and subsequent lack of proper care to many people.
6. The access to mental-health services is limited by geographic and economical barriers; particularly, rural and the population with limited financial resources are limited in accessing mental-health services.

2.0 AI SOLUTION

MindCare AI is a chatbot utilizing Natural Language Processing (NLP) to interpret emotional tone from user input and generate supportive responses.

Key features include:

- A conversational interface for emotional expression.
- AI-based sentiment and emotion classification.
- Supportive responses, coping techniques, and wellness suggestions.
- Alerts or suggestions for additional help based on emotional intensity.

The solution provides a safe, accessible, and private environment for users from diverse backgrounds to express emotions and receive supportive guidance.

3.0 THE GOAL OF AI SOLUTION

- To create a private emotional support platform accessible to all types of users.
- To analyze emotional tone instantly and provide relevant supportive responses.
- To help reduce emotional stress through suggested coping strategies.
- To reduce emotional overwhelm by offering immediate assistance.
- To promote overall mental well-being and encourage healthy emotional regulation.

4.0 PROCESS OF EMPATHIZE IN DESIGN THINKING

To build a successful product, we need to understand our user's needs and goals. We make sure that we are addressing the correct issue rather than just the most apparent one by thoroughly understanding their needs, motivations, and pain spots. This user-centric foundation is critical for creating solutions that are not only functional but also meaningful, relevant, and genuinely valuable to the people we are designing for.

WHO are we empathizing with?

We empathize with individuals privately struggling with common feelings of stress, anxiety and burnout which can stem from professional pressure, academic demands, financial worries, or personal responsibilities. Our empathy focuses on the crucial gap—caused by strong obstacles—between feeling overburdened and asking for assistance. These include practical access issues like high costs or lack of time, and the self-minimizing belief that their problems "aren't severe enough" to justify formal support.

What do they need to DO?

These users need a way to find immediate, accessible, and private validation the moment they feel overwhelmed. They must be able to launch an application on a personal device, freely express their emotions by typing a message, and get an immediate, encouraging answer that acknowledges what they are going through in order to achieve this. This simple action should provide them with a sense of relief, offer guided exercises or relevant resources, and ultimately help them feel understood and less alone in their struggle.

What do they SEE?

The users see overwhelming information about mental health on websites, apps, and social media. They notice that there are limited immediate support options available to them. They are aware of the social stigma surrounding mental health, which prevents them from seeking help openly. They also observe that friends, colleagues, or peers are facing similar emotional challenges, which can increase their sense of isolation.

What do they SAY?

The users often say, "I feel stressed but don't know who to talk to." Some express, "I want help but don't want anyone to judge me." Others mention, "I need something that can help me anytime, not just during appointments." They also frequently express confusion or worry about understanding their own emotional state and how to manage it effectively.

What do they DO?

The users often browse mental health websites or mobile apps to find guidance and coping strategies. They try self-help exercises, such as meditation, journaling, or breathing techniques. They may occasionally discuss their feelings with friends or family, though many avoid professional help due to stigma, cost, or fear of judgment. Without consistent support, they may struggle to manage stress effectively.

What do they HEAR?

The users hear advice from friends or family, such as, “Just relax, it’s not a big deal.” They hear social media and news emphasizing productivity and success, which can increase pressure. They also hear occasional reminders from apps or notifications encouraging them to meditate or take breaks. They may hear peers sharing similar struggles, but rarely receive actionable guidance or professional support.

What do they THINK AND FEEL?

Pain: The users frequently feel isolated and misunderstood, believing their struggles are unique. This is compounded by guilt over their inability to cope and a persistent anxiety about being judged negatively by others.

Gain: The users have a fundamental need for a confidential space where they can feel heard and validated without criticism. They also seek actionable strategies to manage their well-being and a sense of support to overcome their feeling of isolation.

5.0 PROCESS OF DEFINE IN DESIGN THINKING

Based on the empathizing process, we can define the user needs and requirements as follows:

1. An immediate and accessible method for users to express their feelings and get support at any time.
2. A private and non-judgmental platform where users can talk about their emotions without fear of stigma.
3. Automatic understanding and validation of users' emotional states through their text input.
4. Provision of relevant coping strategies and supportive responses based on the user's current emotional needs.
5. Maintenance of complete confidentiality while providing meaningful mental health support.